(CRS) by the Registrar General of India (RGI), which publishes annual reports based on the number of all registered deaths submitted by states and UTs. The report for 2020, published on May 3, showed 8.1 million deaths, a six per cent rise over the previous year, and an excess of 474,806 deaths. But WHO estimates this latter figure to be 820,000. The official Covid toll for that time? Only 149,000.

India objects to the use of Global Health Estimates (GHE) 2019—region- and country-wise data on death and disability—in one of the models used by WHO for calculating excess mortality for India. "Scientists and doctors still don't know the mortality likelihood of Covid. How can a model predict the chances of people dying from this disease, particularly when it is based on another estimate like the GHE," asks a source in the Union health ministry. WHO also used test positivity rates to infer mortality—again, with its inherent range of erraticity, not a solid ground to work from, as India points out.

THE DEA(R)TH OF DATA

But one issue with our official toll is that, unlike in the US or UK, not all deaths are registered. The ministry claims a 99.9 per cent coverage in death registration in India in 2020 (even though news reports indicate a drop in registration that year in states like Telangana, Uttar Pradesh, Delhi and Uttarakhand). That is a significant jump from 2019's 92 per cent and 84.6 per cent in 2018, and the government thus seeks to accord a greater veracity to the CRS figure of 475,000 excess deaths in 2019-2020. "We had 700,000 excess deaths the year before. Without looking at actual data from the country, modelling exercises are pointless," says NITI Aayog member Dr V.K. Paul.

Rebuttals are flowing from the WHO side, even if not officially yet. "Science should be responded to with science. The government should make all national data public and show its alternative estimate," says Bhramar Mukherjee, professor of biostatistics at the University of Michigan School of Public Health and an advisor to WHO on excess mortality. Jon Wakefield, member of WHO's technical advisory group, put out extracts from a paper citing the fact that the estimates are "based on data from India". WHO had "access to data from just 17 of the 36 states and Union territories"—"either reported directly by the states through official reports" or by journalists who obtained official information through RTI requests, the paper said.



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Also, the 2019 report of the Vital Statistics of India-based CRS shows death registration is hardly uniform across states—Tamil Nadu, Kerala, Karnataka, Haryana, Andhra Pradesh, Goa, Odisha and Punjab have 100 per cent registration, but Manipur has only 21 per cent, Nagaland 30 per cent, UP 63.3 per cent and Bihar 51.6 per cent. Furthermore, only 11 states that have over 90 per cent death registration make this data available within 21 days of the death occurring. That's why Covid death data is still being updated for many states—Kerala alone has added 21,000 new deaths for April 2021.

The WHO report is not the first independent assessment on the trail of Covid deaths in India that dwarfs the country's official toll—at least five pre-print and peer-reviewed studies, including in journals like *Science* and *PLOS One*, peg such deaths at over three million till mid-2021. One estimate touched 4.5 million. All these estimates relied on CRS data. There are four key sources of death statistics in India—the CRS, the Sample Registration System (SRS), and indirect estimates from the decennial census and the National Family Health Surveys (NFHS). SRS, a major source of demographic indicators in India, is based on a system of dual recording of births and deaths garnered from field investigation in a sample set of villages and urban blocks—which is then extrapolated for the whole country. The next SRS report, for 2020, may be one way to ascertain if there was undercounting of Covid deaths.

At any rate, the debate frames an area of concern—the poverty of data—that offers vital lessons for India. For any pandemic mapping and, indeed, for all health indices and disease preparedness. "There has to be both accuracy and transparency of data. This pandemic is a signal for India to have a system where death data is collected in entirety, with details on causes," says noted virologist T. Jacob John. Currently, the cause of death is never listed during registration—a basic infirmity. Malini Aisola, co-convenor of AIDAN (All India Drug Action Network) and public health specialist, offers the perfect coda: "One can only improve a health system if we do not hide from the weaknesses." ■